Prabhāsa: Pāṇinian Structured Pretraining for Small Language Models

Anonymous ACL submission

Abstract

Data-efficient pretraining of small language models raises a sharp question: which inductive biases actually improve generalization under a fixed token budget, and which only appear to until they are tested with proper controls? We study this with Prabhāsa, a pre-registered, matched-budget evaluation of whether Pāṇinian morphosyntactic structure—Vidyut morpheme-boundary embeddings, kāraka role-stratified masking, and a supervised role-prediction objective—improves small-model pretraining on the official BabyLM suite. We train a 114M-parameter bidirectional encoder on the Strict (100M-word) and Strict-Small (10M-word) budgets and test every claim with three to five seeds, paired bootstrap inference, and Holm–Bonferroni correction. The main positive finding is a scale-dependent objective effect: pure masked-language modelling (MLM) is statistically indistinguishable from a hybrid MLM + CLM objective at 10M tokens but exceeds it by 5.49 BLiMP points at 100M, which complicates the hybrid recipe used by the BabyLM 2024 winner GPT-BERT. We treat this 100M figure as a single, seed-sensitive estimate: two further seeds, started from identical weights, land 14–19 points lower. Against this, both Pāṇinian mechanisms return pre-registered null or weak results once their confounds are removed: at matched mask budget, kāraka-stratified masking is causally inert; and a kāraka auxiliary objective gives only +0.76 targeted points over five seeds (not significant). On the official BabyLM 2026 harness the submission checkpoints score 67.56 and 59.46 BLiMP against baselines of 74.53 and 65.08; our internal selection harness scores the same checkpoints about five to six points higher, a systematic offset we document. The contribution is a controlled separation of the design choices that help at this scale (objective and architecture) from a linguistically motivated prior that does not, with code, checkpoints, and official prediction files released.

Keywords: data-efficient pretraining; BabyLM; Pāṇinian grammar; kāraka roles; masked language modelling; morphology-aware masking.

1 Introduction

Scaling has driven most recent progress in language modelling, yet it obscures a basic question: under a fixed data budget, which design choices actually buy generalization, and which only seem to until tested with matched controls? Small models trained under the strict budgets of the BabyLM challenge (Warstadt et al., 2023; Hu et al., 2024) are an unusually clean setting to ask this, because the data is held fixed and the confounds that dominate frontier-scale training are absent. What survives is informative: it constrains the design space for data-efficient pretraining and tells us which inductive biases are worth their complexity.

One long-standing source of inductive bias is linguistic theory. Classical Pāṇinian grammar (Matilal, 1990) encodes compositional structure morphologically, through segmentation and inflection, and at the argument level through kāraka categories that assign each participant a semantic role (agent, patient, instrument, recipient, source, locus) defined by its relation to the action rather than by surface order (Kiparsky and Staal, 1969; Cardona, 1974). Because kāraka roles are semantic, they are in principle language-independent, a natural target for a role-aware pretraining signal even in English. The intuition is familiar: a prior that shrinks the effective hypothesis space should reduce the data needed for a given competence. Whether this holds for masked-language-model pretraining, when tested rather than assumed, is our empirical question.

Prabhāsa investigates this through a pre-registered hypothesis hierarchy with two mechanistically distinct levers. RQ-A (masking causality) asks whether, at matched mask-rate budget, stratifying token masking by kāraka role changes downstream syntax, realized via Vidyut morpheme-boundary N-hot embeddings (Prasad, 2024). RQ-B (auxiliary objective) asks whether a supervised token-level kāraka-role prediction head (a computational analogue of śābdabodha, verbal cognition) supplies a signal masking alone does not. Primary evaluation targets the BLiMP agreement and argument-structure subsets, with pre-registered thresholds of at least $+1.0$ point on the targeted subset, assessed across at least three seeds with paired bootstrap resampling and Holm–Bonferroni correction.

The findings run against the usual expectation. The clearest effect is not from the Pāṇinian mechanisms but from the pretraining objective, and it is scale-dependent: neutral at 10M tokens but a 5.49-point pure-MLM advantage at 100M, which relates to GPT-BERT (Charpentier and Samuel, 2024), whose hybrid recipe we find is not uniformly preferable as scale grows (Figure 2). Both Pāṇinian levers return pre-registered null or weak outcomes once their confounds are removed. This is what the controls are for: an earlier single-seed run had suggested a $+1.23$-point gain from kāraka masking, which the matched-budget comparison shows was an artifact of unequal effective mask rates. The contributions are (i) a mechanistic separation of RQ-A and RQ-B under matched budgets with pre-registered inference; (ii) a scale-dependent objective effect connected to the BabyLM state of the art; (iii) an explicit account of training-seed sensitivity at 100M and of a systematic evaluation-harness offset; and (iv) open release of code, checkpoints, and official-pipeline predictions.

2 Related Work

Data-efficient and structure-aware pretraining. The BabyLM challenge (Warstadt et al., 2023; Hu et al., 2024) isolates training methodology from data scale through fixed 10M/100M-token budgets; competitive submissions such as LTG-BERT (Samuel et al., 2023), ELC-BERT (Charpentier and Samuel, 2023), and the 2024 winner GPT-BERT (Charpentier and Samuel, 2024) encode structural priors in masking, architecture, or objective. A recurring lever is the masking distribution: rather than masking tokens uniformly (Devlin et al., 2019), morphology- and curriculum-aware schemes condition masking on linguistic structure (Xu et al., 2020; Belinkov et al., 2017; Botha and Blunsom, 2014). Our masking and auxiliary mechanisms sit in this line but differ in coupling the masking schedule to an explicit semantic-role system. We adopt two now-standard components as the architectural baseline: RoPE positional encoding (Su et al., 2021) and the Muon optimizer (Jordan et al., 2024), and we evaluate causal-versus-masked objective choices in the spirit of Charpentier and Samuel (2024).

Pāṇinian grammar and śābdabodha. Pāṇini's Aṣṭādhyāyī (circa 500 BCE) derives surface forms from abstract stems through ordered rules, with the kāraka relations occupying a meaning-bearing layer that Kiparsky and Staal (1969) connect to the deep–surface distinction of generative grammar and Cardona (1974) formalize as six relation types defined by the participant's role in the action rather than by case or word order. Because kāraka roles are semantic, they are a language-independent target for a role-aware signal even in English. The derivation system is realized computationally by the Vidyut tool-chain (Prasad, 2024). The supervised auxiliary is motivated by śābdabodha, the Navya-Nyāya account of how verbal cognition reconstructs sentence meaning compositionally from word meanings and their kāraka relations (Matilal, 1990; Ganeri, 1999). Prior linguistically motivated pretraining has typically tested a single prior in isolation; our contribution is the integration—kāraka-aware masking and morpheme embeddings tested under pre-registered, matched-budget, multi-seed protocols against the official leaderboard suite.

3 Methods

The core model is a 14-layer Transformer encoder (hidden 768, 12 heads, feed-forward 3072; $\sim$114M / 100M parameters on the 10M / 100M tracks), with RoPE positional encoding (Su et al., 2021) and the Muon optimizer (Jordan et al., 2024). The learning rate follows cosine annealing with 5% warmup; each batch is 256 sequences of length up to 192. Three Pāṇinian mechanisms (Figure 1) are layered on this baseline.

For each token t, Vidyut (Prasad, 2024) produces a morpheme-boundary annotation and a kāraka-role label: one of kartā (agent), karman (patient), karaṇa (instrument), sampradāna (recipient), apādāna (source), and adhikaraṇa (locus), or a separator class for non-role tokens. A sparse binary vector $\mathbf{v}(t) \in \{0,1\}^{10}$ encodes these labels and is

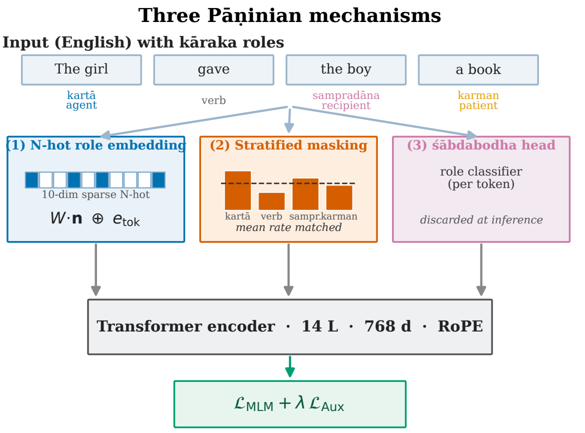

Figure 1: The three Pāṇinian mechanisms. (1) A Vidyut N-hot morpheme/role vector is projected and added to each token embedding; (2) kāraka role-stratified masking sets per-role mask probabilities whose mean is budget-matched to the uniform baseline; (3) a śābdabodha auxiliary head predicts roles from the encoder hidden state and is discarded at inference. The encoder is trained with $\mathcal{L}_{\mathrm{MLM}} + \lambda\,\mathcal{L}_{\mathrm{aux}}$.

projected and added to the standard embedding,

$$\mathbf{x}(t) = \mathrm{Embed}(t) + W\,\mathbf{v}(t), \quad W \in \mathbb{R}^{768\times 10}, \tag{1}$$

so morphosyntactic information enters without any architectural change. Masking adapts by role, $p_{\mathrm{mask}}(t) = \alpha\,p_{\mathrm{base}} + (1-\alpha)\,p_{\mathrm{role}}(r(t))$, with per-role probabilities rescaled to hold the mean mask rate constant; any effect is then attributable to the *distribution* of masked tokens across roles rather than to total masked volume (the budget-matching control central to F2). The auxiliary objective adds a token-level role classifier whose loss is combined with the MLM loss at weight λ (typically 1.0); the head is discarded before evaluation, so no role labels are needed at test time. For English, where Vidyut has no native parser, roles are assigned by a token-level heuristic over a small Sanskrit role lexicon and part-of-speech cues; this introduces label noise that we return to as a limitation.

Two objectives are compared: a hybrid MLM + CLM configuration (15% masked, plus a causal next-token head weighted 0.5) and a pure-MLM configuration (CLM head removed). All models are evaluated on the official BabyLM suite: BLiMP (Warstadt et al., 2020) (67 minimal-pair paradigms), BLiMP Supplement, EWoK (Ivanova et al., 2024), Entity Tracking, COMPS (Misra et al., 2023), and GLUE (Wang et al., 2019). The two mechanism studies (F2, F3) pre-register a *targeted* subset of 20 BLiMP paradigms drawn from

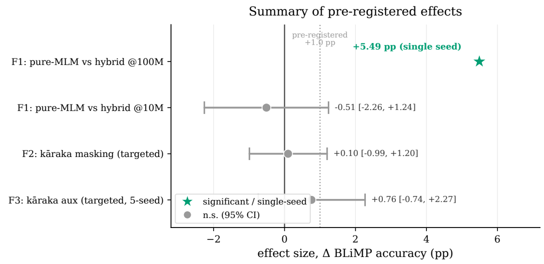

Figure 2: Summary of pre-registered effects on BLiMP (points). The 100M objective effect (F1) is the one large positive, but it is single-seed and seed-sensitive (Figure 4); the 10M objective contrast and both kāraka mechanisms fall within noise of the +1.0-point threshold (dashed). Whiskers are 95% intervals.

the agreement and argument-structure categories, where role information is most likely to matter. Reported results are means with 95% confidence intervals over three to five seeds (single seed for the exploratory 100M arms); arm comparisons use paired bootstrap over per-paradigm subsets with Holm–Bonferroni correction. The Strict baseline is 74.53 BLiMP and Strict-Small 65.08 (Hu et al., 2024). We also ran a frozen pre-pretraining "dose" ablation, injecting synthetic Pāṇinian, Dyck, or English sequences before the main run; it produced no effect at proxy scale (Appendix B) and is not used in the submissions. Corpus, training stages, and the full hyperparameter table are in Appendix A and Appendix B.

4 Results

We report three pre-registered findings: F1 (a scale-dependent objective effect, our principal positive), F2 (kāraka masking at matched budget), and F3 (a kāraka auxiliary objective over five seeds). Figure 2 places all four pre-registered effects on a common axis. Table 1 situates both submission models on the official BabyLM 2026 leaderboard harness against the official baselines.

4.1 F1: Scale-Dependent Objective Effect

A hybrid objective adds a causal language-modelling (CLM) head to the MLM loss, which may stabilize learning when the MLM signal is sparse but dilute it once MLM is strong. Pure-MLM and hybrid models are trained identically except for the objective, with three seeds at 10M tokens and single seeds at 100M. At 10M, pure-MLM achieves a 3-seed mean of 63.58 (95% CI ±1.73) versus hybrid 64.09 (±0.26), with overlapping intervals. At 100M, pure-MLM reaches 73.06 against

Table 1: Official BabyLM 2026 pipeline (the leaderboard harness) for both submission models against the official baselines (Hu et al., 2024); per-task GLUE in Appendix C. Our internal development harness scores the same checkpoints ∼5–6 points higher (73.06 / 64.09 BLiMP), a systematic offset (§4.5). The Strict-Small Entity-Tracking entry (—) is a task that produced no predictions and is scored 0 server-side, not a missing baseline.

	Strict (100M)		Strict-Small	
Task	ours	base	ours	base
BLiMP	67.56	74.53	59.46	65.08
BLiMP Suppl.	63.81	65.00	55.08	57.25
COMPS	53.78	55.85	50.99	51.81
EWoK	52.35	—	52.24	—
Entity Tracking	22.28	23.58	—	21.07
GLUE (macro)	63.28	—	63.09	—

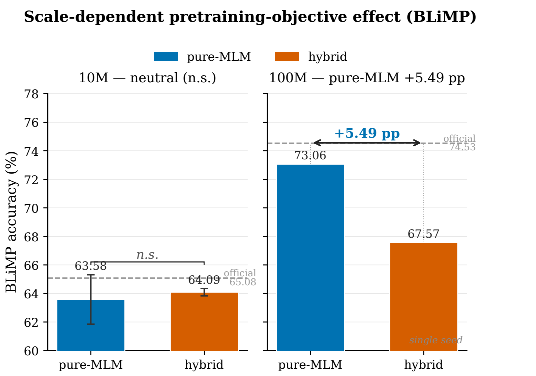

Figure 3: F1: pure-MLM versus hybrid MLM+CLM across the 10M- and 100M-token budgets (internal harness).

hybrid 67.57, a 5.49-point advantage (Figure 3; internal harness). The CLM head's regularizing effect at 10M appears to become a bottleneck at the larger budget. This bears on GPT-BERT (Charpentier and Samuel, 2024), whose hybrid masked-plus-causal training won BabyLM 2024: on our data the hybrid recipe is not uniformly preferable, and at 100M tokens pure MLM is the stronger choice on BLiMP. The 10M neutrality justifies retaining hybrid for the Strict-Small submission; the 100M result drives the Strict-track model.

Training-seed sensitivity at 100M. We report the 100M figures as single-seed estimates. Rerunning the locked recipe from identical initialization under two further seeds does not recover 73.06: best masked-LM losses of 1.61 / 1.82 (versus the submission seed's 0.515) and BLiMP of 58.3 / 53.6, with one optimizer trajectory diverging (Figure 4). Because initialization is held fixed, all

Table 2: F1: BLiMP by objective and budget (internal harness; n=3 at 10M, n=1 at 100M). The objectives are neutral at 10M but diverge by 5.49 points at 100M.

Budget	Objective	BLiMP
10M	Pure-MLM	63.58 (±1.73)
10M	Hybrid	64.09 (±0.26)
100M	Pure-MLM	73.06
100M	Hybrid	67.57

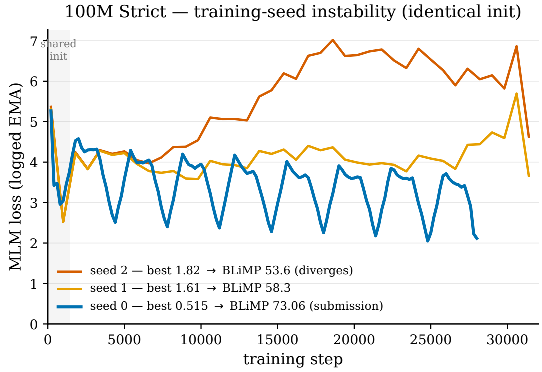

Figure 4: Training-seed instability of the 100M recipe. All three seeds share an identical initialization (shaded) and begin together, then diverge: seed 0 (the submission) descends to the basin scoring 73.06, while seeds 1–2 plateau or climb and score 58.3 / 53.6.

trajectories begin identically and separate only during optimization; the evaluation is not implicated (the submission checkpoint re-evaluates to 73.06 and an independent pseudo-log-likelihood proxy reproduces all three scores within 0.4 points). The 100M recipe is thus not yet robust to the training seed: 73.06 is a valid but fortunate-seed outcome, and the magnitude of the 100M advantage is seed-sensitive. Stabilizing the recipe (gradient clipping, optimizer warmup, best-of-N) is future work.

4.2 F2: Kāraka-Masking Causality (Matched Budget)

Kāraka role-stratified masking (Arm K) is compared to uniform masking (Arm C) at identical mask budget (100M tokens, $n = 1$ per arm), holding N-hot embeddings, RoPE, and pure-MLM fixed and disabling frequency-weighted masking ($\alpha = 0$). The pre-registered primary metric is a paired bootstrap over 20 targeted paradigms. Arm K achieves 71.77 full / 82.03 targeted BLiMP; Arm C achieves 70.49 / 81.93. The targeted contrast is $\Delta_{K-C} = +0.10$ points (95% CI $[-0.99, +1.20]$), not significant. Earlier exploratory runs reported a +1.23-point gain, but that reflected unmatched mask rates and frequency weighting; with matching

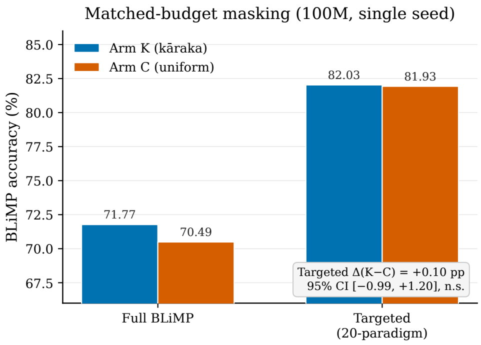

Figure 5: F2: role-stratified masking (Arm K) versus uniform masking (Arm C) at matched mask budget, on full and targeted BLiMP (100M tokens). The targeted gap is +0.10 points, within noise.

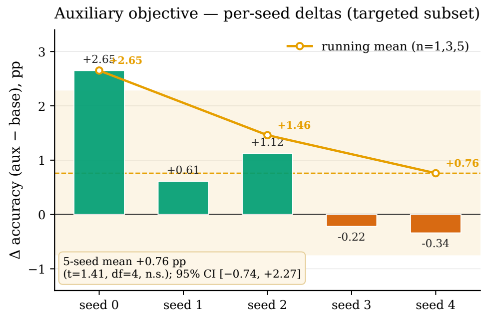

Figure 6: F3: per-seed auxiliary-minus-baseline delta on the targeted subset (10M tokens), with the running mean attenuating from +2.65 (one seed) to +0.76 (five seeds).

Table 3: F3 specificity control (3-seed means, 10M tokens): the real kāraka auxiliary versus a shuffled-role control and a no-auxiliary baseline. The shuffled control does not recover the effect.

Condition	Targeted	Full BLiMP
Real auxiliary	74.41	65.37
Shuffled-role	73.53	63.69
Baseline (no aux.)	72.95	63.74

enforced, the role distribution contributes no signal (Figure 5). A separate 10M arm with structured masking enabled corroborates the null, falling below the hybrid baseline (62.08 versus 64.09). F2 is a pre-registered null; per our closure rule a single-seed null is preliminary pending further interventions (dependency-parsed roles, 3-seed replication), which we leave to future work.

4.3 F3: Kāraka Auxiliary Objective (5-Seed t-Test)

The supervised kāraka-role head is applied at 10M tokens ($\lambda = 1.0$ versus $\lambda = 0.0$), with corpus, hybrid objective, uniform masking, and seed list fixed. The pre-registered primary metric is the per-seed targeted delta. The 5-seed targeted mean is auxiliary 74.02 versus baseline 73.26 (+0.76 points, 95% CI $[-0.74, +2.27]$; $t = 1.41$, df$=$4, n.s.), attenuating from +2.65 at one seed to +1.46 at three to +0.76 at five (Figure 6); the seed-0 result was favourable seed variance, not a stable effect. On full 67-paradigm BLiMP the auxiliary reaches 64.89 versus baseline 63.88 (+1.01, n.s.). A specificity control sharpens the reading (Table 3): a shuffled-role auxiliary, which keeps the role-label distribution but destroys token alignment, does *not* reproduce the real-auxiliary effect (targeted 73.53 versus 74.41), which rules out generic multi-task regularization and is consistent with a weak, kāraka-specific signal that is underpowered at 10M.

4.4 Submission Models and GLUE

On the internal harness, the Strict submission reaches 73.06 BLiMP, within 1.5 points of the 74.53 baseline, and the Strict-Small submission reaches a 3-seed mean of 64.09 (±0.26) against the 65.08 baseline (Table 4). The Strict-Small submission keeps the hybrid objective because F1 shows the pure-MLM advantage only emerges at 100M. Fine-tuned on seven GLUE tasks (Wang et al., 2019), the official-pipeline macro-averages are 63.28 (Strict) and 63.09 (Strict-Small), strongest on MRPC and WSC and weakest on the multi-class task MNLI (Appendix C, Table 7)—the expected profile for these data budgets.

4.5 Two Harnesses, and the Gap Between Them

All controlled comparisons above use a single internal masked pseudo-log-likelihood harness applied identically to every arm. The official BabyLM 2026 pipeline (Table 1) places the same checkpoints ∼5–6 points lower (67.56 versus 73.06 BLiMP for Strict; 59.46 versus 64.09 for Strict-Small). Both harnesses load the identical public checkpoints and the offset is systematic, so this is a scoring-convention difference, not a model difference, and it leaves the relative F1–F3 conclusions intact (they are within-harness contrasts). We report the official numbers as the leaderboard record and use the internal harness only for relative comparison. The practical implication is that a model-selection harness must be calibrated against the scoring harness of record, or absolute claims will not transfer.

Table 4: Submission models on the internal development harness (the basis for all controlled comparisons). Official-pipeline / leaderboard numbers are in Table 1.

Track	Budget	BLiMP	Baseline
Strict	100M	73.06 (n=1)	74.53
Strict-Small	10M	64.09 (±0.26)	65.08

5 Discussion and Conclusion

The reproducible gains come from two sources: positional encoding via RoPE and the choice of pretraining objective, whose effect is strongly scale-dependent (neutral at 10M, a 5.49-point pure-MLM advantage at 100M on the internal harness). In contrast, the kāraka mechanisms carry no measurable causal weight on standard linguistic diagnostics: role-stratified masking is statistically indistinguishable from uniform masking at matched budget (the earlier apparent gain traced to mask-rate confounds), and the auxiliary's five-seed mean attenuates toward zero while resisting a shuffled-role control. These nulls do not falsify the hypothesis that argument structure matters for pretraining; they narrow scope, showing the specific mechanisms tested do not help at the 10M–100M-token budget on English BLiMP. Larger models, longer training, multilingual settings, or compositional-generalization benchmarks (COGS, SCAN, CFQ) (Kim and Linzen, 2020; Lake and Baroni, 2018; Keysers et al., 2020) may behave differently.

The value is methodological as much as empirical: pre-registration with effect-size thresholds, matched-budget controls that remove the mask-rate confound, multi-seed validation that exposes inter-seed variance, and full reporting of nulls, of training-seed instability at 100M, and of a systematic evaluation-harness offset. Two cautions follow: a single fortunate seed can pass for a robust result at 100M, and a selection harness that is not calibrated to the scoring harness of record will overstate absolute performance. Both are documented here. Code, checkpoints, and the official-pipeline prediction files are released (anonymized for review).

Limitations

Several limitations bound these conclusions. **Scale.** The budgets explored (10M and 100M tokens; a 114M-parameter model) are far smaller than contemporary pretraining, and the F1 scale-dependence shows that mechanism efficacy can change with capacity; whether the kāraka nulls hold at a 350M-token budget or above is untested. **Seed robustness.** The 100M recipe is not yet robust to the training seed (§4.5), so the 73.06 result is a fortunate-seed outcome rather than a stable estimate, and several of the 100M arms (F1 hybrid, F2 Arm K/C) are single-seed and therefore preliminary by our own closure rule. **Role-label noise.** For English, kāraka roles come from a token-level heuristic rather than a parser; this injects noise that could mask a real effect, and replacing it with dependency-parsed roles is the most direct intervention for F2 and F3. **Evaluation surface.** The mechanism studies target BLiMP agreement and argument structure; the nulls do not rule out effects on compositional-generalization benchmarks (COGS, SCAN, CFQ) or on Sanskrit-specific evaluation, which we did not run. **Harness.** Absolute numbers depend on the scoring harness; we report both the official pipeline (the leaderboard record) and our internal selection harness, but the ∼5–6 point offset means internal absolute numbers should not be compared across harnesses. **Single language.** Training and evaluation are English-centric despite a Sanskrit-derived prior; cross-lingual transfer, a natural setting for kāraka structure, is left to future work.

Ethics Statement

This work uses the public BabyLM corpora, which carry the biases of their web and book sources; the released models are research artifacts not intended for deployment. The Pāṇinian role inventory is specific to a Sanskrit grammatical tradition, and practitioners adapting the role scheme to other languages should re-validate the mappings rather than assume transfer. No human subjects or private data were involved.

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Appendix A Architecture and Training Configuration

Both tracks use a 14-layer Transformer encoder (hidden 768, 12 heads, feed-forward 3072) with RoPE positional encoding and the Muon (Newton–Schulz orthogonalized momentum) optimizer at peak learning rate 1×10^{-3} with cosine annealing and 5% linear warmup. Batch size is 256 sequences of length up to 192; the masking rate follows a cosine schedule from 0.40 to 0.15 over 10 epochs. The tokenizer is a 20,000-piece SentencePiece BPE built jointly over English and Sanskrit. Both tracks use the same ∼114M-parameter architecture and differ only in the token budget (100M words for Strict, 10M for Strict-Small); each is trained on a single 128 GB unified-memory accelerator, with a 100M-token run taking ≈13.7 hours and a 10M-token run ≈82 minutes. The N-hot projection $W \in \mathbb{R}^{768\times10}$ maps the sparse role vector to the hidden dimension and is added to the token embedding before encoding; the auxiliary head is a token-level role classifier trained with weight λ and discarded at inference. Table 5 lists the full hyperparameters; both pretraining tracks share these settings and differ only in the token budget and (for Strict-Small) the objective.

Table 5: Full pretraining hyperparameters. Both tracks share the architecture and optimization; Strict uses pure MLM at the 100M-word budget, Strict-Small uses the hybrid MLM + CLM objective at 10M words. GLUE fine-tuning uses learning rate 3×10^{-5}, batch size 16–32, sequence length 192, up to 10 epochs (30 for WSC).

Hyperparameter	Value
Layers	14
Hidden size	768
Attention heads	12 (head dim 64)
Feed-forward size	3072
Positional encoding	RoPE
Parameters	∼114M
Tokenizer	SentencePiece BPE
Vocabulary size	20,000 (joint en+sa)
Max sequence length	192
Optimizer	Muon
Peak learning rate	1×10^{-3}
LR schedule	cosine, 5% warmup
Batch size	256 sequences
Epochs	10
Mask-rate schedule	cosine $0.40 \rightarrow 0.15$
Objective (Strict)	pure MLM
Objective (Strict-Small)	hybrid MLM + CLM
CLM head weight (hybrid)	0.5
N-hot role dimension	10
Auxiliary weight λ	1.0 (when enabled)
Frequency weight α (F2)	0.5 / 0 (matched)
Seeds	3 (10M), 1 (100M arms), 5 (F3)
Hardware	1×128 GB unified-mem. GPU

Appendix B Corpus, Curriculum, and Dose Arms

Pretraining uses the official BabyLM Strict (100M words) and Strict-Small (10M words) English corpora mixed with a small in-budget Sanskrit component from the Digital Corpus of Sanskrit and GRETIL, plus a synthetic Pāṇinian dose generated by the Vidyut-Prakyā derivation engine; English comprises ∼95% of tokens, and the total stays within the track word budget. Vidyut annotations (morpheme boundaries, kāraka roles) are precomputed and cached for determinism. A frozen pre-pretraining dose ablation at 60M-token proxy scale compared Arm A (English control), B (Pāṇinian), C (Dyck control), and D (Paribhāṣā), scoring 64.15 / 63.54 / 63.85 / 63.77 BLiMP—a 0.61-point spread (null), so the dose was not used in the leaderboard submissions.

Appendix C Reproducibility and Per-Task Results

Locked recipe. The validated 100M Strict submission was trained at a frozen code state (released in the anonymized repository). Recipe: objective MLM, dose-arm A with zero pre-pretraining epochs, ten English epochs, RoPE, N-hot embeddings, role-stratified masking (cosine 0.40→0.15, frequency weight $\alpha = 0.5$), max sequence length 192, vocabulary 20,000, Muon at peak LR 1×10^{-3}, seed 0; best loss 0.515. Re-running this recipe from identical initialization under two further seeds reaches best losses 1.61 / 1.82 and BLiMP 58.3 / 53.6 (Table 6); the divergence is a training-seed instability the seeding exposed, not an evaluation error (the checkpoint re-evaluates to 73.06 on the official pipeline, and an independent proxy reproduces all three scores within 0.4 points). Code, checkpoints, and official prediction files are released (anonymized for review).

Table 6: Training-seed sensitivity of the 100M recipe (identical initialization).

Seed	best MLM loss	BLiMP
0 (submission)	0.515	73.06
1	1.61	58.3
2	1.82	53.6

Official per-task GLUE. Table 7 reports the official-pipeline GLUE results (zero-shot results are in Table 1).

Table 7: Official 2026-pipeline GLUE (accuracy, except MRPC/QQP F1) for both submission models, with baselines where reported.

Task	Strict	base	S-Small	base
BoolQ	67.03	67.46	65.99	65.87
MNLI	48.84	59.94	46.37	49.80
MRPC	82.04	84.35	82.57	83.49
MultiRC	57.55	63.90	57.88	64.52
QQP	64.83	70.73	63.03	60.86
RTE	61.15	56.83	60.43	60.43
WSC	61.54	—	65.38	—
Macro	63.28	—	63.09	—

F1/F3 per-seed (internal harness). F1 10M pure-MLM seeds: 65.22 / 63.31 / 62.20 (mean 63.58); hybrid 3-seed mean 64.09 ($\pm$0.26). F3 targeted per-seed deltas: $+2.65, +0.61, +1.12, -0.22, -0.34$ (5-seed mean $+0.76$, $t = 1.41$, df$=$4, n.s.).